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Smart Blood Donation Management Platform (Hayat)

منصة ذكية لإدارة التبرع بالدم (حياة)

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ABSTRACT

In emergency healthcare environments and resource-constrained regions, such as the Gaza Strip, the prompt availability of blood units is often hindered not by a deficit of willing donors, but by fragmented communication, reliance on unorganized social media appeals, and lack of geographical coordination. To overcome these logistical barriers, this project presents Hayat, a centralized, resilient web-based blood donation coordination platform designed to optimize donor management and emergency response workflows. By integrating geolocation filtering, targeted donor matching, multi-channel alert mechanisms (including Web Push and SMS fallbacks), and Progressive Web App (PWA) offline capabilities, Hayat minimizes matching latency and ensures system availability during network disruptions. Operating strictly within a communication and logistical scope, the platform bridges the gap between verified blood repositories and eligible voluntary donors without interfering with clinical laboratory procedures. Ultimately, Hayat provides an efficient, scalable software solution aimed at saving lives by ensuring timely, organized, and targeted donor mobilization during urgent medical crises.

DEDICATION

We dedicate this project to our families, whose continuous support, encouragement, and sacrifices have been the foundation of our academic journey.

To our respected teachers and professors, who provided us with knowledge, guidance, and inspiration throughout our years of study.

To every voluntary blood donor whose generosity gives hope and saves lives, and to the healthcare workers and blood bank personnel who continue to carry out their humanitarian mission under difficult circumstances.

Finally, we dedicate this work to the people of the Gaza Strip, whose patience, resilience, and determination have taught us that hope and dedication can overcome even the greatest challenges.

ACKNOWLEDGEMENTS

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Our appreciation also goes to all the individuals who participated in the requirements elicitation questionnaire and contributed their opinions and experiences. Their responses helped us understand the real challenges associated with blood donation coordination and guided the development of the Hayat platform.

Finally, we would like to thank our families, friends, and everyone who supported, encouraged, or assisted us during the preparation and completion of this project.

Table of Contents

CHAPTER 1: INTRODUCTION 1
1.1 Project Background 1
1.2 Problem Statement 2
1.3 Project Objectives 3
1.3.1 Primary Objective 3
1.3.2 Specific Objectives 3
1.4 Project Scope 4
1.4.1 In-Scope (Functional & Technical Boundaries) 4
1.4.2 Out-of-Scope (Explicit System Exclusions) 4
CHAPTER 2: LITERATURE REVIEW AND RELATED WORK 6
2.1 Introduction 6
2.2 The Role of Information Technology in Blood Donation Coordination 6
2.3 Review of Existing Regional Systems 7
2.3.1 Wateen Application 8
2.3.2 Wareed Application 9
2.3.3 Arab Electronic Blood Bank Application 9
2.4 Comparative Analysis and Research Gaps 10
2.4.1 Limitations that recur across every system reviewed 11
2.4.2 Gaps specific to infrastructure and coordination design 12
2.5 Theoretical and Technical Foundations 12
2.6 Proposed Approach: Value Proposition and Verification 14
2.6.1 Verification of feasibility 15
CHAPTER 3: RESEARCH METHODOLOGY 16

3.1 Introduction 16
3.2 Research Methodology 16
3.2.1 Research Approach 16
3.2.2 System Development Methodology 16
3.3 Requirements Elicitation Survey 17
3.3.1 Survey Methodology 17
3.3.2 Key Findings and Implications for Requirements 18
3.4 System Analysis 21
3.4.1 Stakeholder Identification 21
3.4.2 System Overview 22
3.4.3 Functional Requirements 22
3.4.4 Non-Functional Requirements 23
3.5 System Modeling 24
3.5.1 Actors Identification 24
3.5.2 Use Case Identification 24
3.5.3 Use Case Diagrams & Detailed Specifications 25
Detailed Use Case Specifications (Tables) 25
3.5.4 Entity Relationship Diagram (ERD) 29
REFERENCES 31

List Of Figures

) Figure 3.1: Respondents' Awareness of Their Own Blood Type..... 18

) Figure 3.2: Prior Blood Donation Experience 18

) Figure 3.3: Current Channels for Receiving Urgent Blood-Donation Appeals..... 19

) Figure 3.4: Perceived Adequacy of Social-Media-Based Appeals..... 19

) Figure 3.5: Preferred Notification Channel During a Nearby Emergency 20

) Figure 3.6: Interest in a Personal Donor Dashboard..... 20

) Figure 3.7: Overall Use Case Diagram 25

List Of Tables

) Table 2.1: Comparison of Selected Blood Donation Coordination Systems 11

) Table 3.1: Summary of Survey Findings and Requirements Implications 21

) Table 3.3 Functional Requirements 23

) Table 3.4 Non-Functional Requirements 23

) Table 3.5 UC-01: Register Account 25

) Table 3.6 UC-02: Login 26

) Table 3.7 UC-03: Manage Profile 26

) Table 3.8 UC-04: View Eligibility 26

) Table 3.9 UC-05: Receive Notification 27

) Table 3.10 UC-06: Respond to Request 27

) Table 3.11 UC-07: View Donation History 27

) Table 3.12 UC-08: Create Blood Request 28

) Table 3.13 UC-09: Manage Blood Requests 28

) Table 3.14 UC-10: Search Eligible Donors 28

) Table 3.15 UC-11: Manage Users 29

CHAPTER 1: INTRODUCTION

1.1 Project Background

Blood donation is universally recognized as a foundational pillar in the advancement of modern healthcare services. The success of critical medical interventions—ranging from emergency trauma care and complex surgical procedures to the long-term management of chronic conditions—is fundamentally contingent upon the immediate availability of safe blood units.

Despite its critical importance, maintaining a consistent blood supply remains a global challenge, particularly in regions grappling with resource scarcity and protracted emergencies [1]. This logistical strain is further compounded by the biological constraints of blood products, which possess a strictly limited shelf life. Consequently, the entire lifecycle of blood collection, storage, and distribution is exceptionally time-sensitive. Inefficient coordination between blood repositories and voluntary donors directly threatens patient outcomes, often resulting in critical shortages during peak demands or tragic wastage due to expired stock.

In recent decades, information technology has emerged as a transformative force in mitigating these healthcare logistical challenges. The integration of centralized databases, automated scheduling, and electronic records has proven vital in minimizing operational delays, enhancing data accuracy, and facilitating rapid, life-saving decision-making [2],[3] .

Conventional methods for recruiting donors during urgent shortages—such as broad public appeals via social media channels and broadcasting—often suffer from low response efficiency, lack of geographic targeting, and severe time latency. Although social media serves as a rapid broadcasting tool, it lacks structured data processing, precise donor-compatibility filtering, and real-time location matching, making it inadequate for critical medical emergencies.

In the Gaza Strip, the chronic shortage of blood units is uniquely characterized; it is primarily driven not by a lack of willing donors, but rather by systemic fragmentation, ineffective communication channels, and a reliance on outdated, traditional coordination methods. To bridge this operational gap, a

specialized digital platform capable of facilitating rapid, targeted communication is urgently required.

To address these localized challenges, a tailored, lightweight coordination platform—Hayat—is proposed for this volatile environment. It is crucial to define the operational scope of such a system: it is strictly a logistical and communication-driven coordination platform. It does not automate clinical laboratory procedures or medical testing. All blood testing, screening, and actual transfusion workflows remain entirely within the manual and clinical jurisdiction of certified blood banks, while Hayat serves as the digital bridge that optimizes the connection between those who need blood and those who are ready to give it.

1.2 Problem Statement

Blood banks frequently face critical shortages of specific blood types during emergencies, not due to an absolute scarcity of donors, but owing to a fundamental lack of coordination and efficient communication. The core problem is the absence of a centralized, automated system to manage donor information and streamline urgent requests.

Currently, communication heavily relies on slow manual procedures or unorganized public appeals via social media platforms. While social media is currently utilized as a rapid broadcasting tool, it lacks targeted geolocation matching and blood-type filtering, leading to chaotic response patterns, delayed matching, and uncoordinated donor mobilization.

According to global health reports, this coordination gap severely hinders the ability to link eligible donors with urgent requests in a timely manner [1]. As a result, critical medical interventions and blood transfusions are delayed, placing patients' lives at immediate risk. This project addresses this challenge through the development of Hayat, an efficient, centralized coordination platform designed to optimize communication, drastically reduce response times, and ensure blood availability during emergencies.

1.3 Project Objectives

1.3.1 Primary Objective

To design and develop a centralized, lightweight web platform (Hayat) that streamlines donor coordination, accelerates emergency response times, and optimizes communication between blood repositories and potential donors in resource-constrained environments.

1.3.2 Specific Objectives

Centralize Donor & Request Management: Create a secure, automated database system to register, categorize, and maintain updated profiles of eligible blood donors based on blood types and eligibility criteria.

Enable Location-Based & Targeted Matching: Implement geolocation filtering mechanisms to identify and notify the nearest suitable donors during emergency blood requests, minimizing latency.

Optimize Urgent Alert System: Develop a multi-channel notification mechanism (incorporating Web Notifications and SMS fallbacks) to ensure reliable reachability even under unstable network conditions.

Streamline Request & Appointment Workflows: Build intuitive dashboards for blood bank administrators to dispatch urgent calls and manage donor responses efficiently without interfering with clinical laboratory procedures.

Ensure Resilience & Offline Capabilities: Leverage Progressive Web Application (PWA) technologies to allow essential data caching and local accessibility during frequent internet disruptions.

1.4 Project Scope

1.4.1 In-Scope (Functional & Technical Boundaries)

Donor Management Module: User registration, profile authentication, blood type management, and tracking last-donation dates to determine eligibility automatically.

Emergency Request & Dispatch System: Administrative tools for verified blood bank personnel to create urgent blood requests filtered by blood type, quantity, and urgency level.

Location-Based Matching & Notifications: Geolocation filtering algorithms to direct emergency requests to nearby eligible donors, supported by Web Push Notifications and SMS fallback mechanisms.

Administrative & Analytical Dashboards: Interactive control panels for blood banks to monitor active requests, track donor responses, and generate basic logistical reports.

Resilient Front-End Interface: Implementation of Progressive Web App (PWA) capabilities to ensure offline accessibility to essential operational data during internet disruptions.

1.4.2 Out-of-Scope (Explicit System Exclusions)

Clinical & Medical Testing Operations: Automated laboratory testing, blood screening, viral disease testing, or clinical compatibility cross-matching. All biological validations remain under the manual jurisdiction of certified healthcare facilities.

Direct Blood Transfusion & Inventory Tracking: Managing physical blood bank storage units, refrigeration parameters, or hospital-to-hospital blood bag transfers.

In-App Financial Transactions: Processing any payments, monetary rewards, or commercial transactions for blood donation.

Hardware & Sensor Integrations: Integration with medical equipment, wearable health trackers, or IoT laboratory hardware.

CHAPTER 2: LITERATURE REVIEW AND RELATED WORK

2.1 Introduction

Blood transfusion medicine occupies an unforgiving position within the healthcare system: it cannot be stockpiled indefinitely, it cannot be manufactured on demand, and it cannot tolerate the latency of manual coordination once an emergency has begun. A safe, adequate blood supply depends entirely on the timely convergence of three elements — a willing donor, a verified need, and a communication channel fast enough to connect the two before the clinical window closes. Where any one of these elements is delayed, the cost is measured not in inconvenience but in patient outcomes.

This chapter examines how digital systems have attempted to close that gap, and where they continue to fall short. The discussion proceeds from the general to the specific: it opens with the role of information technology in blood supply logistics, narrows into a critical examination of three deployed regional platforms, consolidates their shortcomings into a defined set of research and operational gaps, and closes by establishing the theoretical foundations and design commitments that the proposed Hayat platform uses to respond to those gaps. Each subsection is written to build directly on the one before it, so that the platform's design choices in later chapters read as a deliberate answer to a documented problem rather than an arbitrary set of features.

2.2 The Role of Information Technology in Blood Donation Coordination

The shift from paper registers and word-of-mouth appeals toward structured information systems has been one of the most consequential developments in blood-service logistics over the past two decades. Centralized databases, automated donor registries, and internet-connected alerting mechanisms have measurably reduced the operational delay between the moment a shortage is identified and the moment a

matching donor is reached [1]. Web-based coordination platforms in particular have demonstrated that consolidating donor records and emergency requests into a single, queryable system shortens response time and improves donor engagement relative to fragmented, manual alternatives [2].

Yet the same body of work is consistent on a second point: most deployed systems were built for communication volume rather than for coordination precision. They excel at reaching many people quickly but are far weaker at reaching the right people — donors of the correct blood type, within a workable distance, who are actually eligible to donate at that moment. Systematic reviews of emergency healthcare logistics identify this as a structural weakness rather than an implementation detail: broad public broadcasting substitutes for, rather than complements, genuine geographic and clinical targeting [3]. The literature is equally clear that any system intended for volatile, infrastructure-constrained regions must be designed around bandwidth limitations, intermittent connectivity, and high-stress operating conditions from the outset, not retrofitted for them after launch [1] [3] [4]. That constraint is the lens through which the three regional systems below are evaluated.

2.3 Review of Existing Regional Systems

Three deployed regional applications were selected for close examination because, between them, they represent the dominant design patterns currently used for donor coordination in the Arab region: a government-backed national platform (Wateen), a donor-registry search tool (Wareed), and a cross-border aggregator (Arab Electronic Blood Bank). Examining all three exposes not a single point failure but a pattern that recurs regardless of who built the system or which country it serves.

2.3.1 Wateen Application

Wateen is a Saudi Arabian mobile application backed by the Ministry of Health, built to formalize the link between voluntary donors and official blood banks. It gives donors visibility into nearby blood bank locations and operating information, and rewards sustained participation through a certification mechanism [4].

Strengths:

- Issues digital certificates of appreciation after repeated donations, giving the platform a retention mechanism that most competitors lack.
- Integrates barcode scanning to track individual donation history with reasonable accuracy.
- Surfaces operating hours and location details for accredited blood banks in one place.

Structural weaknesses:

- The interface becomes a bottleneck precisely when it matters most — during urgent request searches, where users report friction rather than speed.
- The platform exhibits instability and crashes under concentrated load, which is exactly the load pattern an emergency alert generates.
- Operating-hour notifications are inconsistent and delayed, undermining the one feature the app is built around.

2.3.2 Wareed Application

Wareed, developed in Yemen, takes a different approach: rather than broadcasting alerts, it exposes a searchable registry of donors that patients and healthcare providers can query directly to find a compatible blood type [5].

Strengths:

- Donor contact information is protected by genuinely strong privacy controls — the strongest of the three systems reviewed here.
- Registration is simple enough that onboarding friction is unlikely to be a major barrier to donor sign-up.
- The interface is clean and easy to navigate for its core search function.

Structural weaknesses:

- There is no spatial or map-based view of donor distribution, so proximity — the single most operationally relevant variable in an emergency — is invisible to the user.
- Search cannot be constrained to a defined radius or administrative district, which forces manual filtering of results that should be filtered automatically.
- The registry has almost no connection to blood bank inventory or appointment scheduling, so a successful search still ends in an off-platform handoff.

2.3.3 Arab Electronic Blood Bank Application

The Arab Electronic Blood Bank operates at a different scale entirely, connecting patients with volunteer donors across multiple countries rather than within a single national system [6].

Strengths:

• Basic filtering and ranking of donor search results give users at least some control over relevance.

• Cross-platform availability on both Android and iOS removes one common adoption barrier.

Structural weaknesses:

• An outdated interface directly damages both usability and perceived trustworthiness of the platform.

• Donor contact data is exposed with minimal protection — the weakest privacy posture among the three systems reviewed.

• There is no educational content on donor eligibility, leaving screening almost entirely to manual clinical judgment after the fact.

2.4 Comparative Analysis and Research Gaps

Read individually, each system's weaknesses could be dismissed as an implementation flaw specific to that product. Read together, they form a pattern: every reviewed system trades away either reach, precision, resilience, or privacy — and no single system holds all four simultaneously. Table 2.1 makes that trade-off explicit by scoring each platform, including the proposed Hayat design, against the criteria that determine whether a coordination system actually functions under emergency conditions rather than merely existing.

Criterion	Wateen	Wareed	Arab Electronic Blood Bank	Hayat (Proposed)
Deployment model	Native mobile app	Native mobile app	Native mobile app (Android/iOS)	Progressive Web App (browser-based, no install)
Offline / low-connectivity support	No	No	No	Yes — service-worker caching, offline read/write queue
Geolocation-based donor matching	No	No	Basic filtering only	Yes — radius-based filtering around request location
Multi-channel alert delivery	Push notifications only	No push/SMS integration	No SMS fallback	Web Push + SMS fallback
Role-based access control	Limited (donor-facing only)	Limited	Not specified	Yes — donor, blood bank admin, platform admin roles
Donor privacy protection	Not specified	Strong	Weak — contact data exposed	Yes — contact details shielded from public view
Targeted vs. broadcast alerts	Broadcast	Search-driven, not proactive	Broadcast / ranked list	Targeted, location- and type-filtered dispatch
Donation tracking / incentives	Yes — certificates, barcode scanning	No	No	Basic eligibility tracking (last-donation date)

(1) Table 2.1: Comparison of Selected Blood Donation Coordination Systems

2.4.1 Limitations that recur across every system reviewed

Three weaknesses are not unique to any single platform — they appear, in some form, in all three, which is what elevates them from product complaints to a genuine research gap:

• No real-time visibility: none of the three systems gives a blood bank an automated, live view of donor availability or blood-type coverage; shortages are discovered reactively, not anticipated.

• No automated alert pipeline: reaching a donor still depends on that donor actively opening the app or on a human making a phone call — there is no system-initiated push that fires the moment a request is logged.

• Weak donor-side engagement: donors are treated as a searchable resource rather than as users with their own dashboard, eligibility status, and appointment history to manage.

2.4.2 Gaps specific to infrastructure and coordination design

The infrastructure and network vulnerability gap: every native mobile application reviewed here assumes continuous, high-speed connectivity and app-store distribution as a precondition for use. In precisely the environments where blood shortages are most acute — regions with frequent power and network disruption — that precondition fails, and the platform fails with it.

The targeted coordination gap: the systems reviewed split into two unsatisfying camps. Some broadcast urgent requests to the widest possible audience, which produces donor fatigue and chaotic, uncoordinated overcrowding at blood banks. Others restrict access behind closed institutional systems, which solves the chaos problem by sacrificing reach entirely. No reviewed system combines wide public visibility with precise, automatic geographic and type-based filtering, or with a redundant delivery channel that survives a connectivity failure.

2.5 Theoretical and Technical Foundations

The architecture proposed for Hayat is not an arbitrary combination of trending technologies; each design commitment below responds directly to one of the gaps identified in Section 2.4, and each is grounded in an established software engineering principle:

• Centralized information systems: consolidating donor profiles and active requests into a single logical data repository is the direct countermeasure to the fragmentation that undermines Wareed's registry model and Wateen's notification reliability — a single source of truth eliminates the reconciliation errors that fragmented records inevitably produce [2], [7].

• Client–server and web architecture: separating the user-facing interface from backend business logic — implemented as a centralized web application on a PHP/Laravel backend — allows donors, blood bank staff, and administrators to operate against the same system concurrently without one group's load degrading another's experience, which directly addresses the stability failures observed under load in Wateen.

• Role-based access control (RBAC): the exposed-data problem seen in the Arab Electronic Blood Bank is a direct consequence of treating all users as equivalent. RBAC enforces that blood bank administrators hold the elevated privileges required to dispatch emergency calls, while donors are confined to a personal, non-privileged dashboard — privacy by structural design rather than by policy alone [8].

• Event-driven and notification architecture: the absence of an automated alert pipeline, identified as a cross-cutting weakness in Section 2.4, is answered by treating every logged emergency request as a trigger event. That trigger immediately initiates location-based donor filtering and dispatches Web Push notifications with an SMS fallback to eligible donors, removing the dependency on a donor proactively checking an app [9].

• Progressive Web Application (PWA) principles: the infrastructure vulnerability gap is a connectivity problem, not a feature problem, and it is addressed at the architecture level rather than through app-store logistics. Service workers and client-side caching keep essential read and write functions operable through network disruption, which is the specific resilience property native app-store distribution cannot guarantee.

2.6 Proposed Approach: Value Proposition and Verification

The platform developed in response to the gaps above is referred to consistently as Hayat. Its value proposition is best understood not as a list of features but as a direct rebuttal to each weakness documented in this chapter:

- Where native apps fail under connectivity loss, Hayat is delivered as a Progressive Web App, requiring no app-store installation and remaining reachable from any modern browser, desktop or mobile.

- Where alerting depends on manual outreach or a single channel, Hayat dispatches Web Push notifications with an automated SMS fallback, so an urgent request still reaches donors when internet connectivity fails.

- Where geographic targeting is absent or crude, Hayat filters donors within a precise radius of the requesting facility, while keeping personal contact information shielded from public exposure.

- Where donor-side tools are an afterthought, Hayat screens donor eligibility automatically from blood type and last-donation date, and folds appointment scheduling into the same platform rather than leaving it to an off-platform handoff.

Taken together, these commitments position Hayat as the first system among those reviewed to combine offline-capable delivery, precise geographic matching, dual-channel dispatch, and structural privacy protection within one lightweight platform — which is exactly the combination the research gaps in Section 2.4 call for, rather than a partial answer to one gap at the expense of another.

2.6.1 Verification of feasibility

At this stage, feasibility is established analytically rather than empirically: the claim is that the architecture is buildable and internally consistent, not yet that a running system has been measured. That claim rests on a concrete, testable implementation path — a Laravel (PHP) backend serving a PWA front end — broken into milestones that can each be verified independently: schema migrations for donor and request data, authentication and role enforcement, geolocation-based matching logic, and the notification dispatch pipeline itself. Each milestone is a falsifiable checkpoint rather than an assumption, and their successful execution, documented in the methodology and implementation chapters, constitutes the practical evidence that the design proposed here functions as intended rather than merely on paper

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

The primary objective of this chapter is to present the research methodology and system design framework for the Hayat platform—a Progressive Web Application (PWA) designed to streamline emergency blood donation coordination. Developing an efficient healthcare system requires a rigorous analytical approach to understand user expectations, eliminate communication bottlenecks, and structure system architecture effectively.

This chapter outlines the research approach, data collection methods (including empirical findings from a 62-participant study), and the software development methodology adopted. Furthermore, it details the comprehensive system analysis—covering stakeholder identification, system overview, functional/non-functional requirements—and presents formal system modeling, including actor specifications, use cases, detailed use case tables, and database structure.

3.2 Research Methodology

3.2.1 Research Approach

This project utilizes a quantitative, data-driven research approach coupled with an applied engineering method. Quantitative research allows the gathering of measurable insights from potential system end-users. By collecting data directly from the target population, design decisions regarding platform features, alerting channels, and interface mechanisms are based on statistical evidence rather than theoretical assumptions.

3.2.2 System Development Methodology

The system is developed using the Agile Development Methodology (Scrum). Agile was chosen due to its iterative nature, flexibility in handling shifting user

requirements, and emphasis on continuous integration.

The development process was divided into bi-weekly sprints focusing on:

* Sprint 1: Requirement Gathering & Database Modeling.
* Sprint 2: Authentication & Role-Based Access Control (RBAC).
* Sprint 3: Core Request Dispatching & Matching Logic.
* Sprint 4: PWA Offline Capabilities, Service Workers & Dual Notification Engine (SMS/Push).
* Sprint 5: Testing, Evaluation, and Refinement.

3.3 Requirements Elicitation Survey

To ground the requirements developed in this chapter in direct evidence rather than assumption alone, a structured online questionnaire was distributed to residents of the Gaza Strip. The questionnaire probed personal blood-type awareness, prior donation behavior, and, most importantly, the channels through which urgent blood-donation appeals currently reach potential donors. Sixty-two valid responses were collected, and the results below are used throughout this chapter to justify specific functional and non-functional requirements.

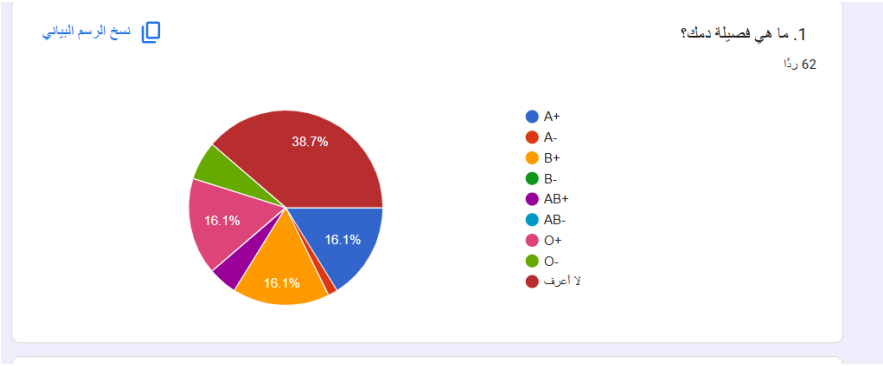
3.3.1 Survey Methodology

The questionnaire was distributed electronically through Google Forms and shared primarily via social media channels, reaching 62 respondents. It consisted of six close-ended questions covering: personal blood-type awareness, prior donation history, the channels currently used to receive urgent blood-donation appeals, the perceived adequacy of those channels, the preferred notification channel during a genuine local emergency, and interest in a personal dashboard for tracking donation eligibility. Because the sample was gathered through convenience sampling rather than probabilistic sampling, the results below should be read as indicative rather than statistically representative of the wider population. They are, however,

consistent with — and reinforce — the coordination gap identified independently in the literature review (Section 2.4).

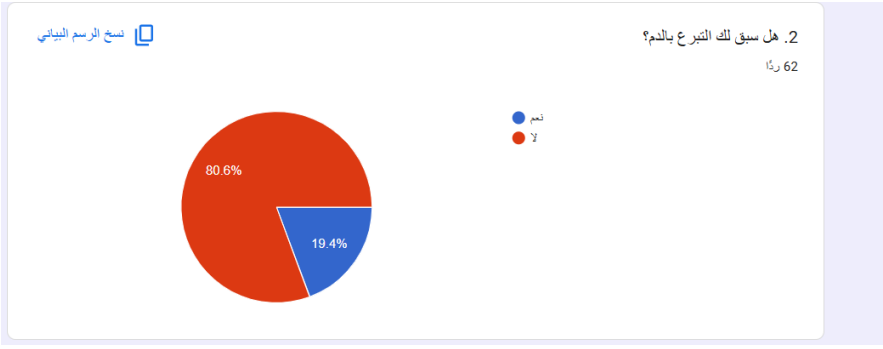
3.3.2 Key Findings and Implications for Requirements

The six figures below summarize the raw survey results; the interpretation following each figure links the finding to the specific requirement(s) it justifies, and Table 3.1 consolidates this mapping.



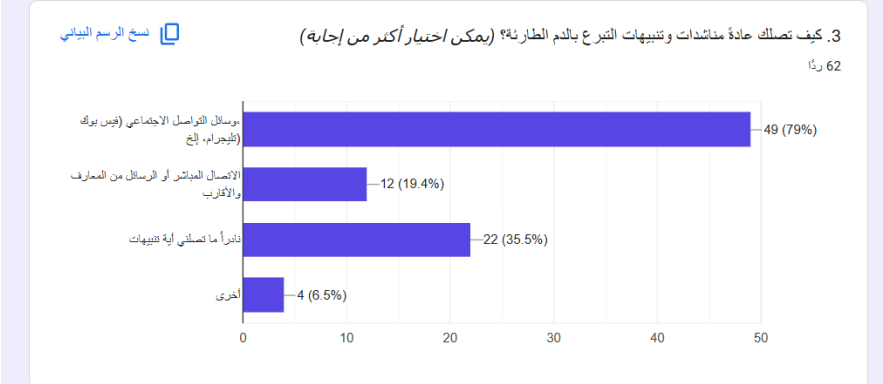
(1) Figure 3.1: Respondents' Awareness of Their Own Blood Type

Only 16.1% of respondents each identified as A+, B+, or O+, while a smaller share reported AB+; the largest single group, 38.7%, selected “I don't know.” Since more than a third of potential donors cannot state their own blood type, the donor registration workflow cannot assume this information arrives ready-formed and must accommodate it explicitly (FR-01 and FR-03).



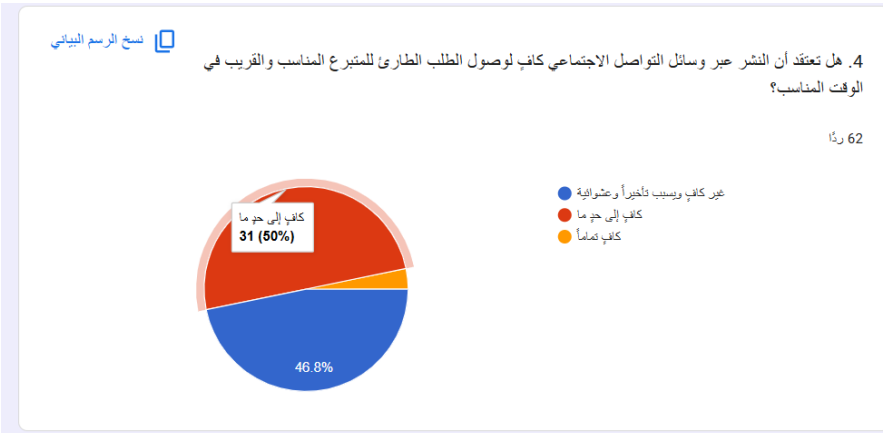
(2) Figure 3.2: Prior Blood Donation Experience

80.6% of respondents had never donated blood before, against 19.4% who had. The platform's primary user base is therefore likely to be first-time donors rather than experienced ones, reinforcing the requirement that the interface remain usable by people with limited prior exposure to formal donation processes (NFR-04).



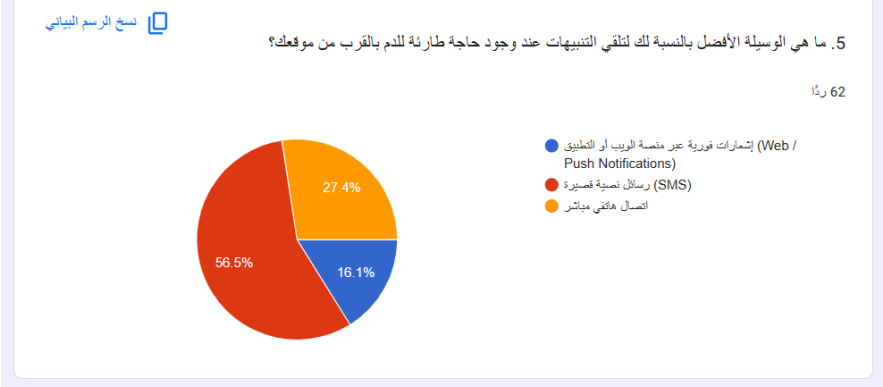
(3) Figure 3.3: Current Channels for Receiving Urgent Blood-Donation Appeals

Respondents could select more than one channel. 79% currently rely on social media platforms for urgent appeals, while 35.5% reported that notifications rarely reach them through any channel at all, and only 19.4% hear about requests through direct contact from acquaintances or relatives. This is direct empirical confirmation of the unstructured, broadcast-only pattern described in Section 1.1, and is the strongest single justification for a targeted, multi-channel dispatch mechanism rather than continued reliance on organic sharing (FR-07 and FR-08).



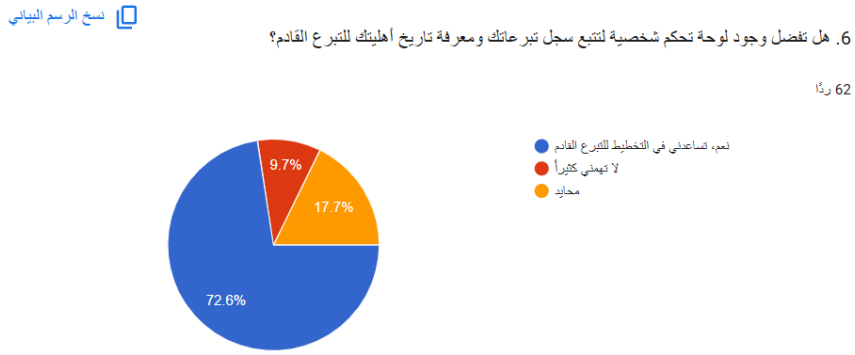
(4) Figure 3.4: Perceived Adequacy of Social-Media-Based Appeals

46.8% of respondents rated current social-media-based appeals as “not adequate, causing delay and randomness,” and 50% rated them as only “adequate to some extent” — together, 96.8% of respondents see the status quo as, at best, partially adequate. This finding directly validates the problem statement in Section 1.2 and the broader rationale for a centralized coordination platform.



(5) Figure 3.5: Preferred Notification Channel During a Nearby Emergency

When asked which channel they would most trust to reach them during a genuine nearby emergency, 56.5% of respondents preferred SMS, 27.4% preferred a direct phone call, and only 16.1% preferred Web or in-app push notifications. This majority preference for SMS reinforces the need for reliable SMS delivery, not merely as a fallback of last resort but as a channel that should be dispatched with equal priority to Web Push in low-connectivity conditions (FR-07).



(6) Figure 3.6: Interest in a Personal Donor Dashboard

72.6% of respondents indicated that a personal dashboard tracking their donation record and eligibility would help them plan future donations, against

17.7% who were neutral and 9.7% who were not interested. This strong majority directly justifies the donor-facing profile, eligibility-tracking, and history features specified later in this chapter (FR-03 and FR-04).

Question	Key Result	Requirement(s) Addressed
Q1 — Blood-type awareness	38.7% of respondents did not know their own blood type.	FR-1, FR-2
Q2 — Prior donation experience	80.6% had never donated blood before.	NFR-6
Q3 — Current alert channels	79% rely on social media for urgent appeals; 35.5% said they rarely receive any notification at all.	FR-8, FR-9, FR-10, FR-11
Q4 — Perceived adequacy of current channels	96.8% rated current channels as only partly adequate or not adequate at all.	Problem Statement (Section 1.2); overall project rationale
Q5 — Preferred channel during a nearby emergency	56.5% preferred SMS; 27.4% preferred a direct phone call; 16.1% preferred Web/push notifications.	FR-10, FR-11
Q6 — Interest in a personal donor dashboard	72.6% wanted a personal dashboard to track donations and eligibility.	FR-2, FR-3, FR-4

(2) Table 3.1: Summary of Survey Findings and Requirements Implications

3.4 System Analysis

3.4.1 Stakeholder Identification

Stakeholders in the Hayat platform are classified into two main categories:

- Primary Stakeholders:

  * Donors: Community members who register to receive targeted emergency alerts, respond to donation appeals, and monitor eligibility.

  * Blood Bank Administrators: Authorized medical personnel responsible for creating, managing, and fulfilling urgent blood requests.

  * System Administrators: Technical administrators responsible for managing

platform entities, users, and overall system security.
- Secondary Stakeholders: Healthcare facilities, patients' families receiving timely donations, and regulatory health bodies.

3.4.2 System Overview

Hayat is a Progressive Web Application (PWA) that acts as a centralized bridge between blood banks and eligible blood donors. When an urgent blood need arises, blood bank administrators log a structured request. The system automatically filters eligible donors within a defined geographical radius based on blood compatibility. Notifications are dispatched instantly via Web Push, with an automated fallback to SMS for offline users. Donors can respond in real time, view hospital location guidance, and track their donation history.

3.4.3 Functional Requirements

ID	Requirement Description	Priority
FR-01	The system shall allow new donors to register with personal, contact, and blood type details.	High
FR-02	The system shall authenticate users securely using JWT/Session control and enforce RBAC.	High
FR-03	The system shall enable donors to update their profiles and geographical locations.	High
FR-04	The system shall calculate and display donor donation eligibility based on historical records.	High
FR-05	The system shall allow Blood Bank Admins to post urgent blood requests specifying units, urgency, and search radius.	High
FR-06	The system shall automatically filter and match compatible donors in proximity to the blood bank.	High
FR-07	The system shall dispatch multi-channel alerts (Web Push Notification with SMS Fallback).	High

ID	Requirement Description	Priority
FR-08	The system shall enable alerted donors to accept or decline incoming emergency requests.	High
FR-09	The system shall allow Blood Bank Admins to manage, track, and close active blood requests.	Medium
FR-10	The system shall allow System Admins to manage platform user accounts and institutional settings.	Medium

(3) Table 3.3 Functional Requirements

3.4.4 Non-Functional Requirements

ID	Attribute	Description
NFR-01	Performance	The platform shall process query matching and alert dispatches within < 3 seconds.
NFR-02	Availability	The platform shall ensure 99.9% operational uptime for emergency request creation.
NFR-03	Security	Passwords must be hashed using bcrypt; sensitive data transmission must be encrypted via HTTPS.
NFR-04	Usability	The user interface must be fully responsive, intuitive, and compliant with accessibility standards.
NFR-05	Offline Support	The PWA service worker must cache static assets to allow basic offline functionality.
NFR-06	Scalability	The backend database structure must support scalable concurrent user requests during critical alerts.

4) Table 3.4 Non-Functional Requirements

3.5 System Modeling

3.5.1 Actors Identification

The system identifies three key operational actors:

1. Donor: An individual user who manages personal eligibility, receives targeted alerts based on location/blood type, accepts requests, and views personal donation history.
2. Blood Bank Administrator: A verified healthcare administrator authorized to initiate emergency blood requests, search local eligible donors, monitor real-time responder counts, and close resolved requests.
3. System Administrator: An administrative authority with system-wide privileges to manage user roles, oversee registered blood bank entities, monitor system health, and generate reports.

3.5.2 Use Case Identification

The core functionalities identified in the system model include:

*UC-01: Register Account

*UC-02: Login

*UC-03: Manage Profile

*UC-04: View Eligibility

*UC-05: Receive Notification

*UC-06: Respond to Request

*UC-07: View Donation History

*UC-08: Create Blood Request

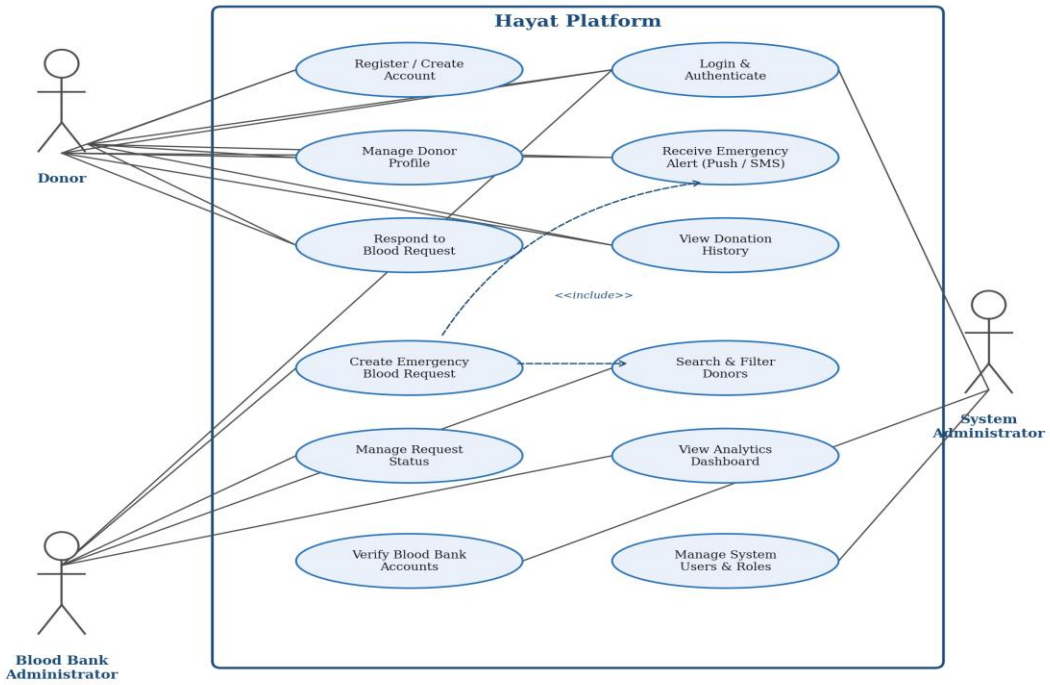
*UC-09: Manage Blood Requests

*UC-10: Search Eligible Donors

* UC-11: Manage Users

3.5.3 Use Case Diagrams & Detailed Specifications

Overall Use Case Diagram:



(7) Figure 3.7: Overall Use Case Diagram

Detailed Use Case Specifications (Tables)

Use Case Name	**UC-01: Register Account**
Actor	Donor
Description	Enables new users to create a donor profile by providing personal details, phone number, location, and blood type.
Precondition	The phone number/email must not be previously registered.
Postcondition	The donor profile is created, saved in the database, and the user is logged in.

(5) Table 3.5 UC-01: Register Account

Use Case Name	**UC-02: Login**
Actor	All Users (Donor, Blood Bank Admin, System Admin)
Description	Authenticates registered users and redirects them to their role-specific dashboard.
Precondition	User must have an active registered account.
Postcondition	User identity is verified, session token is assigned, and access is granted.

(6) Table 3.6 UC-02: Login

Use Case Name	**UC-03: Manage Profile**
Actor	Donor
Description	Allows donors to view and update contact information, address, and notification preferences.
Precondition	Donor must be authenticated.
Postcondition	Database updates the donor profile record.

(7) Table 3.7 UC-03: Manage Profile

Use Case Name	**UC-04: View Eligibility**
Actor	Donor
Description	Automatically evaluates donor eligibility based on historical donation timestamps.
Precondition	Donor must be logged in.
Postcondition	System displays current eligibility status and countdown timer to next eligible date.

(8) Table 3.8 UC-04: View Eligibility

Use Case Name	UC-05: Receive Notification
Actor	Donor
Description	Delivers emergency request alerts to targeted donors based on location and blood type.
Precondition	An emergency request matching donor parameters is posted by a blood bank.
Postcondition	Notification is delivered to the donor's device (Web Push with SMS Fallback).

(9) Table 3.9 UC-05: Receive Notification

Use Case Name	UC-06: Respond to Request
Actor	Donor
Description	Enables alerted donors to confirm or decline availability for an urgent request.
Precondition	Donor receives an emergency alert and is eligible to donate.
Postcondition	Response status is logged and blood bank dashboard counter updates in real time.

(10) Table 3.10 UC-06: Respond to Request

Use Case Name	UC-07: View Donation History
Actor	Donor
Description	Displays historical records of all completed donations made by the user.
Precondition	Donor is authenticated.
Postcondition	Chronological donation history list is rendered.

(11) Table 3.11 UC-07: View Donation History

Use Case Name	**UC-08: Create Blood Request**
Actor	Blood Bank Administrator
Description	Allows blood bank staff to broadcast an emergency request for specific blood types.
Precondition	Blood Bank Admin is authenticated.
Postcondition	Request is published, stored, and matching pipeline is executed.

(12) Table 3.12 UC-08: Create Blood Request

Use Case Name	**UC-09: Manage Blood Requests**
Actor	Blood Bank Administrator
Description	Enables blood bank admins to monitor incoming donors, edit parameters, or close active requests.
Precondition	Active request exists in system.
Postcondition	Request status is updated (Active / Fulfilled / Cancelled).

(13) Table 3.13 UC-09: Manage Blood Requests

Use Case Name	**UC-10: Search Eligible Donors**
Actor	Blood Bank Administrator
Description	Provides manual search/filter tool to locate active eligible donors.
Precondition	Admin is authenticated.
Postcondition	List of matching donors is presented.

(14) Table 3.14 UC-10: Search Eligible Donors

Use Case Name	UC-11: Manage Users
Actor	System Administrator
Description	Provides system admin control over accounts, roles, and administrative access.
Precondition	Authenticated System Admin access.
Postcondition	Account roles or statuses are updated.

(15) Table 3.15 UC-11: Manage Users

3.5.4 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) defines the logical database structure of the Hayat platform. The architecture consists of core entities connected through relational keys:

1. USERS: Stores fundamental user data (id, name, phone, email, password, role_id, city, latitude, longitude, blood_type_id).
2. ROLES: Defines permissions (Donor, Blood Bank Admin, System Admin).
3. BLOOD_TYPES: Reference table storing standard blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-).
4. BLOOD_BANKS: Stores hospital/institution profiles (id, name, city, contact_number, latitude, longitude).
5. BLOOD_REQUESTS: Contains emergency request details (id, blood_bank_id, blood_type_id, units_needed, units_fulfilled, urgency_level, status, search_radius).
6. DONATIONS / RESPONSES: Junction table tracking donor interactions with requests (id, user_id, request_id, response_status, donation_date).
7. NOTIFICATIONS: Logs outgoing push/SMS alerts (id, user_id, request_id, channel, status, sent_at).

Structural ERD Map:

* USERS (1) —— (N) RESPONSES: One donor can respond to multiple requests.

* BLOOD_BANKS (1) —— (N) BLOOD_REQUESTS: One blood bank can post multiple emergency requests.
* BLOOD_REQUESTS (1) —— (N) RESPONSES: One emergency request receives multiple donor responses.
* BLOOD_TYPES (1) —— (N) USERS / BLOOD_REQUESTS: Referenced across donors and requests for automated matching.

REFERENCES

[1] World Health Organization, "Blood safety and availability," WHO Fact Sheets, 2023.

[2] M. Alam, M. S. Hossain, and A. Rahman, "Design and implementation of a real-time web-based blood donation platform," IEEE Access, vol. 8, pp. 142100–142112, 2020.

[3] A. Kumar and P. Singh, "Centralized information systems for emergency healthcare logistics: A systematic review," IEEE Transactions on Engineering Management, vol. 68, no. 4, pp. 1020–1032, Nov. 2021.

[4] Saudi Ministry of Health, "Wateen platform for blood donation management," Riyadh, Saudi Arabia, Tech. Rep., 2022.

[5] Yemen Ministry of Health, "Wareed digital blood donor registry," Sana'a, Yemen, Tech. Rep., 2021.

[6] Arab Electronic Blood Bank, "Cross-regional blood donation initiative report," Cairo, Egypt, Tech. Rep., 2020.

[7] I. Sommerville, Software Engineering, 10th ed. Boston, MA, USA: Pearson, 2016.

[8] R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 9th ed. New York, NY, USA: McGraw-Hill Education, 2020.

[9] S. Hussain and M. Abdullah, "Event-driven notification architectures in emergency response applications," in Proc. IEEE Int. Conf. Softw. Eng. Service Sci. (ICSESS), Beijing, China, 2019, pp. 245–249.